INCH-POUND

MIL-P-63074A (AR)
AMENDMENT 5
12 June 1995
SUPERSEDING
AMENDMENT 4
15 JUNE 1990

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M120
LOADING ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-63074A (AR) dated 17 December 1979, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 SPECIFICATIONS, MILITARY, add:

"MIL-A-70625 - Automated Acceptance Inspection Equipment Design, Testing and Approval, of."

PAGE 2

3.3.1: Delete in its entirety and substitute:

"Ignition element charge composition. The moisture content of the ignition element charge composition after drying shall not exceed 0.02 percent."

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* 3.8.1: Delete in its entirety and substitute:

"3.8.1 Head loading assembly. The Retainer and Closing Plug Loading Assembly shall comply with drawing torque requirements. (See 4.6.7.1)".

* Add 3.8.2:

"3.8.2 Primer loading assembly. The Primer Head/Body threaded joint shall comply with drawing torque requirements. (See 4.6.7.2)".

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* 4.4.2.6: Delete in its entirety and substitute new paragraph 4.4.2.6.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Head Loading Assembly	1 OF 1	9327345
			NEXT HIGHER ASSEMBLY: 9296617

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Non-functioning test		4.5.4	3.5.2	4.6.9
Major					
101	Closing disk missing or damaged exposing explosive		0.40%	3.2	Visual or gage
102	Retainer torque failure		0.40%	3.8.1	4.6.7.1
103	Retainer lacquer or stake missing	1/	0.40%	3.2	Visual
104	Functioning test		4.5.3	3.5.1	4.6.8
105	Electrical resistance test		0.40%	3.4	4.6.3
106	Depth to electrode		0.40%	3.2	Gage
107	Flash hole obstructed		0.40%	3.2	Gage or Visual
108	Closing plug torque failure		0.40%	3.8.1	4.6.7.1
109	Lacquer missing from closing plug threads	1/	0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: Inspections requiring access to inner components and inspections conducted by automatic assembly machine may be performed before completion of entire Head Loading Assembly.

1/ For automatic machine assembly, disassemble 1 unit/hr. and visually inspect. Absence of lacquer/stake or inadequate lacquer application will cause rejection of all production since last acceptable sample was inspected.

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4.4.2.7: Note 1: Delete in its entirety and substitute:

"Three samples minimum shall be selected from each drying oven after completion of drying cycle. If the moisture content of any sample exceeds the requirement, that quantity of material represented by the sample shall either be redried until acceptable results are achieved or rejected. The selected samples must not be dried more than any other unit which could be selected."

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* 4.4.2.8:

Major 107: Delete "4.6.7" and substitute "4.6.7.2".

Critical Defect: Add:

"2 Plug missing 100% 3.2 Visual".

PAGE 19.1

* 4.4.2.14: Delete in its entirety.

PAGE 21

4.6.2.1: Delete in its entirety and substitute:

"Igniter charge composition. The moisture content of the igniter charge composition shall be determined as specified in Method 102.2.1 of MIL-STD-1234 except that the electric ignition element loading assembly and only a two hour dry time shall be used. The upper insulator from the electric ignition element loading assembly may be removed prior to analysis."

4.6.2.1.1: Delete in its entirety.

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* 4.6.7.1: Delete in its entirety and substitute:

"4.6.7.1 Head loading assembly. Torque shall be verified by either performing torque inspections of assembled components or by periodic verification of the torque machine. When torque is verified by inspection of assembled components, torque tests shall be performed by reapplication of torque, while the lacquer is still wet (within 5 minutes of application), and with the part suitably marked and held for detection of movement between adjacent parts. Any discernible movement between adjacent parts is a torque test failure. When torque application is confirmed by monitoring the torque machine, a periodic verification of machine output values is required. Detailed requirements of the verification technique shall be in accordance with MIL-A-70625A and 4.5.2. Failure to meet the requirement will cause rejection of all production since the last successful verification."

* Add 4.6.7.2:

"4.6.7.2 Primer loading assembly. After a 2 hour minimum curing of sealant, the torque shall be applied in the loosening direction, while holding the head. The joint under test shall be marked to provide a means of detecting any movement between adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure."

PAGE 23

4.6.10, 4.6.10.1, 4.6.10.2, 4.6.10.3, 4.6.10.4: Delete in their entirety and substitute the following:

"4.6.10 Determination of ignition charge composition. The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following percentages:

Component	Percentage
Potassium Chlorate and Gum Solids	46.1 ± 2.1%
Lead Thiocyanate	35.5 ± 2.0%
Charcoal	20.0 ± 1.5%

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4.6.10.1 Charcoal. Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven 70°C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml portions of a 5 percent potassium nitrate solution followed by 50 ml of distilled water. Use a small glass rod to bread up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 ± 5°C for 2 hours, cool in a desiccator and weigh. Calculate the percentage of charcoal as follows:

$$\text{Percent charcoal} = \frac{(A-B)\ (100)}{A}$$

A = Beginning sample weight
B = Empty crucible weight

4.6.10.2 Lead thiocyanate. Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid (Nitric acid used should be water white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent lead thiocyanate} = \frac{16.17\ (AB-CD)}{E}$$

A = Volume of silver nitrate solution used
B = Normality of silver nitrate solution
C = Volume of potassium thiocyanate solution
D = Normality of potassium thiocyanate
E = Weight of sample

4.6.10.3 Potassium chlorate & gum solids. The percentage of potassium chlorate and gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent."

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4.6.11: Delete in its entirety and substitute:

"4.6.11 Proving ground test. The primers shall be assembled into complete rounds for which the primer is standard and fired from an M60 or M60A1 tank (or from any appropriate facility mount) with an M68 gun in accordance with ATP IS-P-7J. The test shall be fired with a standard service charge at 145°F -5°F at resultant pressures."

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